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# Effect of aerobic exercise on physical performance in patients with Alzheimer's disease

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Q4 **Abstract** ■■■

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Q5 *Keywords:* ■■■

## 1. Introduction

Alzheimer's disease (AD) is characterized by a progressive decline in cognitive and physical function leading to loss of independence and consequently institutionalization. The disease is costly for the caregivers, communities, and the health care system. So far, pharmacological treatment has proven limited effect, and increasingly more attention is given to nonpharmacological interventions to preserve function and quality of life in community-dwelling patients with AD [1].

Several studies have suggested beneficial effects of different types of exercise interventions on various outcomes of physical function and activity of daily living in community-dwelling patients with cognitive impairments [2–7]. However, most studies have included heterogeneous groups of patients, and sample sizes have been small, which limits the generalizability of the findings [2,3,5,8]. One large study of 210 home-dwelling patients with AD demonstrated a positive effect of a 12-month exercise program on the deterioration of physical function evaluated by the caregivers but was unable to identify any effects regarding objective measures of physical function [6].

The exercise mode (aerobic exercise, resistance exercise, balance training, functional training, and dual-task training), duration, and intensity of exercise interventions may affect specific types and stages of dementia differently. This emphasizes the need for large-scale studies with standardized exercise protocols in well-characterized populations.

Impairments related to performing a motor-task and a cognitive task simultaneously (dual-task performance) are found in early stages of AD, and this is associated with functional decline, disability, and an increased risk of falls [9]. Despite these important consequences, only few studies

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have investigated the effect of exercise on dual-task performance in patients with all-cause dementia [10–12].

Numerous studies have shown that self-efficacy, defined as an individual's belief in his/her ability to complete a specific task, influences a broad range of health behaviors and is linked to levels of physical activity, physical function, and well-being in older adults [13]. One study including older adults with memory complaints showed higher levels of exercise self-efficacy after 6 months of exercise [14]. However, to our knowledge, no studies have investigated the effect of exercise on exercise self-efficacy in patients with AD.

This study includes secondary outcomes from a large-scale randomized controlled trial with the aim to investigate the effects of moderate-to-high–intensity aerobic exercise with cognition as the primary outcome [15]. The aim of the present study was to investigate the effect of moderate-to-high–intensity aerobic exercise on single-task physical performance, dual-task performance, and exercise self-efficacy in community-dwelling patients with mild AD. In addition, this study investigates the association between results in tests of physical performance, cognition, and neuropsychiatric symptoms.

## 2. Materials and methods

The ADEX study (Preserving Cognition, Quality of Life, Physical Health, and Functional Ability in Alzheimer's Disease: the Effect of Physical Exercise study) was designed as a multicenter single-blinded RCT including participants from January 2012 to June 2014. The rationale, design, and the results regarding the effects of the physical exercise intervention on cognition, health-related quality of life, activities of daily living, behavioral, and psychological symptoms have been published previously [15,16].

This article presents specific effects of physical exercise on single-task physical performance, dual-task performance, and exercise self-efficacy. In addition, the article presents the associations between outcomes of physical performance and self-efficacy and (1) the Symbol Digit Modalities Test (SDMT), which is the primary cognitive outcome of the ADEX study and (2) the 12-item Neuropsychiatric Inventory (NPI), because we found positive effects on neuropsychiatric symptoms (NPI) in the ADEX study [15].

The study was preceded by an uncontrolled pilot study to ensure feasibility and safety as suggested by the Medical Research Council [17,18]. The ADEX study (Clinical Trials.gov no.: NCT01681602) was approved by The Danish National Committee on Biomedical Research Ethics (H-3-2011-128) and the Danish Data protection Agency (J NO. 30-0718). Before inclusion in the trial, written informed consent was obtained from each participant.

### 2.1. Participants and study procedures

Participants were recruited consecutively from 8 memory clinics in Denmark [16]. Eligibility for inclusion was (1) a diagnosis of AD according to the NINCDS-ADRDA criteria [19], (2) an MMSE score of ≥20, (3) age between 50–90 years, (4) a caregiver who was willing to participate in the study and who were in contact with the participant more than once monthly, and (5) if receiving anti-dementia or mood stabilizing medication, the dose should be stable for at least 3 months before inclusion. Exclusion criteria were (1) co-morbidity that prevented exercising including musculoskeletal, joint, neurological, and cardiac diseases and (2) performing regular moderate-to-high–intensity exercise twice or more weekly before the study entry. A detailed description of the inclusion and exclusion criteria has been published previously [16]. At each memory clinic, participants were subsequently randomized in blocks of 4–10 participants to an intervention group or a control group using a computerized random-number generator in favor of the exercise program if an odd number of participants were included.

### 2.2. Exercise intervention

Before starting the exercise, the participants in the IG and their caregivers were invited to an information meeting regarding the exercise program and how to prevent overuse injuries.

The IG conducted exercise 1 hour, 3 times weekly for 16 weeks in groups of 2–5 participants supervised by a qualified physical therapist. All exercise sessions included a general warm up and cool down period. The first 4 weeks focused on adaptation to exercising and included strength training of the lower extremity muscles and an introduction to aerobic exercise. The following 12 weeks included moderate-to-high–intensity aerobic exercise on ergometer bicycle, cross trainer, and treadmill in 3 periods of 10 minutes with 2–5 minutes pause in-between. The aerobic exercise was individually tailored and planned to achieve an intensity of 70%–80% of maximal hazard ratio (HR; 220 minus the person's age). To ensure training at the intended intensity, the heart rate was monitored during the aerobic exercise, which included 3 × 10 minutes exercise and pauses (in total 34–40 minutes), and the average heart rate was registered in a training logbook. The average exercise intensity for each individual was defined as: average HR of all sessions/maximal HR (%). In patients who were treated with beta-blockers, the Borg Scale of Perceived Exertion was used to ensure that the intended training intensity was achieved [20]. The attendance rate was defined as: total number of attended exercise sessions/total number of offered exercise sessions (%). Participants were not excluded due to low adherence. All trainers completed a two-day course to ensure standardization and high quality of the training intervention and recording of data.

Participants assigned to the CG received usual care during the 16-week intervention period. Usual care included planned outpatient visits (every 6 months) and contact to the memory clinic staff if necessary due to medication or other needs.

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### 2.3. Assessments

Experienced physical therapists and nurses performed assessments of physical performance and neuropsychologists or specially trained memory clinic staff members assessed cognition. All assessors completed a 1-day course to ensure standardization and high quality of the test procedures and were blinded to allocation and prior test results of the participants.

#### 2.3.1. Test of physical performance

The 6-min Astrand Cycle Ergometer test (Monark Ergomedic 839E; Monark Exercise AB, Sweden) is a submaximal test estimating the maximal oxygen uptake based on workload and average heart rate during the last minute of the 6-minute cycle test, corrected for age and body weight. The participant's heart rate should stabilize in the range of 110 to 170 beats/min in the last minute of cycling to be considered a valid test [21,22]. If a valid test was not achieved at baseline or at follow-up, the data were handled as missing values as described in the statistical analysis.

Timed Up & Go test (TUG) assesses basic mobility by measuring the time it takes for a person to stand up from a chair, walk 3 meters, cross a line, turn around, and walk back to the chair and sit down. The participants were given instructions to complete the test as quickly and safely as possible [23–25].

The 30-s chair stand test (STS) assesses strength and endurance in the lower extremity by measuring the number of stands completed in 30 seconds with hands crossed against the chest [25,26].

The 400-m walk test (fast gait speed) was used to assess walking endurance. The walking time(s) was measured on a 20-meter course marked with two colored cones [25,27,28].

The 10-m walk test (usual gait speed) assesses usual gait speed (m/s) on a 10-m long course [10,25,29].

#### 2.3.2. Test of dual-task performance

Two different dual-task performances were assessed by combining the 10 m walk test with a test of cognitive performance, that is, (1) naming the months backward starting with January (dual-task month) with the command "Walk at your usual gait speed while naming the months backwards starting with January", and (2) counting backward from 50 by ones (dual-task numbers) with the command "Walk at your usual gait speed while counting backwards from 50". The participants were not instructed to prioritize neither the gait nor the cognitive task [10,25,29].

#### 2.3.3. Test of exercise self-efficacy

Exercise self-efficacy was assessed by a five-item questionnaire (score range, 5–25) developed to differentiate between the confidence of perceived ability to participate in exercise despite barriers as bad mood, bad weather, or tiredness [14,30].

#### 2.3.4. Tests of cognition and neuropsychiatric symptoms

The SDMT is used to assess mental speed and attention by a simple substitution task. The participant is asked to identify and pair nine different symbols corresponding to the numbers 1 through 9 for 120 seconds [31].

The 12-item NPI assesses behavior and symptoms common in dementia, including delusions, hallucinations, agitation, depression, anxiety, disinhibition, irritability, and apathy based on interviews with caregivers [32].

### 2.4. Statistical analysis

Wilcoxon nonparametric signed rank test for continuous variables and Fisher exact test for categorical variables were used to compare baseline data and outcome measures at baseline between the IG and the GC.

The difference in change in outcomes from baseline to follow-up at 16 weeks between randomization groups was analyzed in linear regression models using generalized estimating equations to account for repeated measurement. To account for possible clustering by training groups, clinic and group indicator variables were included in the model. The measurements that were available at follow-up were weighted by the inverse of an estimate of the probability of still being in the study [33]. These probabilities were estimated from the data in a logistic regression model with the patient's baseline characteristics and the observed outcome at baseline as covariates.

Besides an intention-to-treat analysis (ITT), we performed a per-protocol analysis excluding participants from the IG with an attendance of <66% of offered exercise sessions. This attendance threshold was based on the evidence-based recommendation of the American College of Sports Medicine [34].

To investigate the association between changes in SDMT, NPI, and all outcomes of single-task physical performance, dual-task performance, and self-efficacy, we used Spearman rank correlation.

The level of significance was set at $P \leq .05$.

## 3. Results

Of 608 pre-screened for eligibility, 200 participants were recruited to the study and randomized. The dropout rate from allocation to the assessments at 16 weeks was 4% in the IG and 8% in the CG. The enrollment, the allocation process, and reasons for dropout are presented in Fig. 1.

Valid results in the Astrand Cycle Ergometer test were achieved in 91 (85%) in the IG and 71 (76%) in the CG. Reasons for not achieving the targeted heart rate of 110–170 BPM was primarily use of beta-blockers or other medication preventing the increase in heart rate when exercising, and in a few of the participants joint pain or other comorbidities, which made it too hard to keep the resistance for 6 minutes.

The participants (n = 200) had a mean age of 70.5 ± 7.4 years and an average MMSE score of

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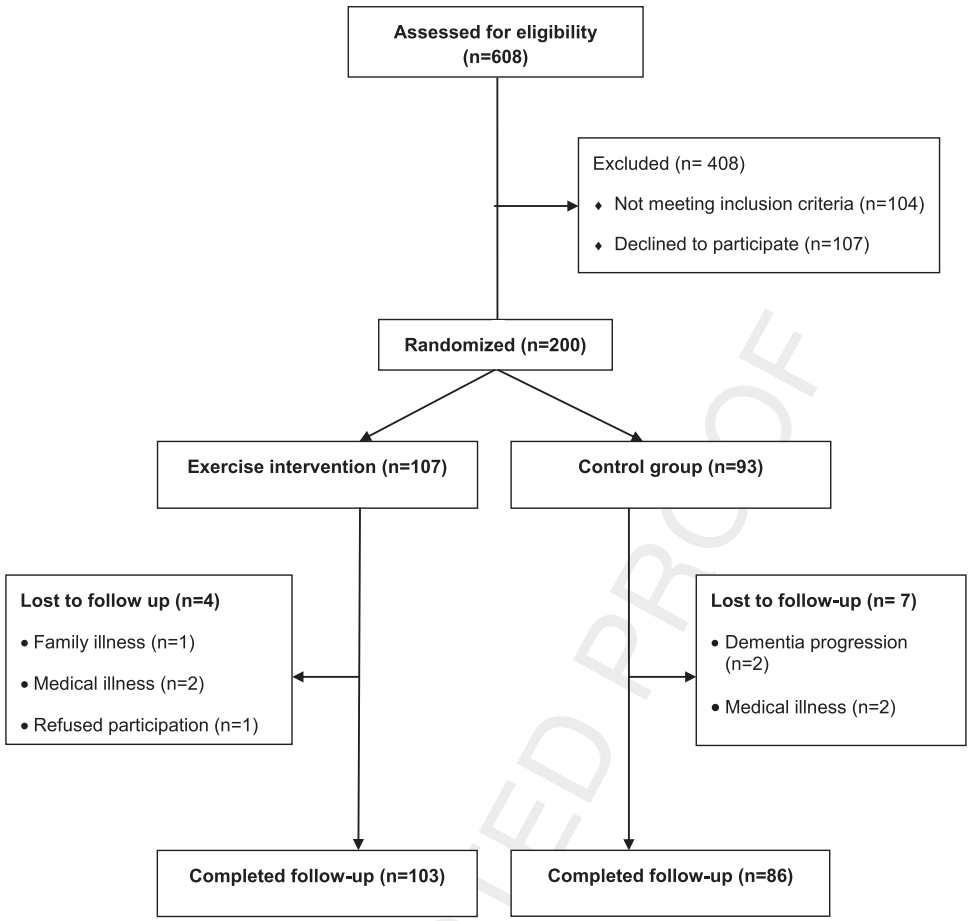


Fig. 1. Flowchart of the ADEX study.

24.0 ± 3.6. There were no significant between-group differences at baseline except that the IG walked significantly slower (m/s) compared to the CG in both tests of dual-task performance and in the 10-m walk test (Table 1).

### 3.1. Attendance, intensity, and safety

Attendance rate was 83.7% (19.5%; mean [SD]) in the IG during the 16 weeks intervention period with 91 (85%) participants attending 66.6% or more of all the offered exercise sessions. Moreover, the exercise intensity was moderate-to-high in the IG during the 12 weeks of aerobic exercise with an average of 81.3% (9.7%) of maximal heart rate for participants not using beta-blockers. Training was safe with only one serious adverse event (atrial fibrillation) possibly related to the exercise intervention. Musculoskeletal problems (11%) and dizziness/faintness (10%) were the most commonly reported adverse events in the IG but only half of those were related to the intervention [16].

### 3.2. Effects of exercise intervention

#### 3.2.1. Intention-to-treat analyses

Significant between-group differences in favor of the IG were found for estimated maximal oxygen uptake and for exercise self-efficacy. All between-group differences in change from baseline to 16 weeks are shown in Table 2.

There were no significant between-group differences in change from baseline in the remaining tests of single-task physical performance or dual-task performance. However, the results were numerically in favor of IG for TUG, 10-m walk, 400-m walk, dual-task month, and dual-task numbers.

#### 3.2.2. Per-protocol analyses (exercise attendance ≥ 66.6% in the IG versus CG)

The between-group differences for estimated maximal oxygen uptake and exercise self-efficacy remained significant after the intervention in the per-protocol analysis. In addition, significant between-group differences in favor of the IG was found for TUG, 10-m walk, 400-m walk, and dual-task numbers indicating a positive effect for exercise attendance ≥66.6% regarding single-task physical performance and dual-task performance (Table 3).

### 3.3. Associations

Equal positive correlations were found between dual-task numbers (m/s) and SDMT in the ITT analyses (rho = 0.21, $P$ = .039) and in the per-protocol analyses (rho = 0.20, $P$ = .048). The analyses revealed no other significant

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Table 1
Q8 Baseline characteristics

| | Intervention group | Control group | *P* value |
|---|---|---|---|
| *n* (%) | 107 (53.5) | 93 (46.5) | |
| Gender (men), *n* (%) | 56 (52.3) | 57 (61.3) | .253 |
| Age (y), mean (SD) | 69.8 (7.4) | 71.3 (7.3) | .163 |
| Caregiver living with patient, *n* (%) | 78 (72.9) | 66 (71.0) | .875 |
| Comorbidities, *n* (%) | | | |
| Hypertension | 48 (44.9) | 35 (37.6) | .317 |
| Diabetes | 11 (10.3) | 7 (7.5) | .623 |
| Hypercholesterolemia | 40 (37.4) | 34 (36.6) | .904 |
| Apoplexia | 1 (0.9) | 2 (2.2) | .598 |
| Acute myocardial infarction | 2 (1.9) | 1 (1.1) | .645 |
| Medicine, *n* (%) | | | |
| Hypertension medicines | 48 (44.9) | 39 (41.9) | .776 |
| Depression medicines | 29 (27.1) | 28 (30.1) | .641 |
| Alzheimer medicines | 105 (98.1) | 88 (94.6) | .254 |
| Beta-blockers | 7 (6.5) | 9 (9.7) | .445 |
| Cognitive performance, mean (SD) | | | |
| MMSE score | 23.8 (3.4) | 24.1 (3.8) | .526 |
| SDMT score | 27.1 (14.7) | 25.4 (14.3) | .405 |
| NPI score | 10.0 (10.8) | 9.4 (9.7) | .699 |
| Physical performance, mean (SD) | | | |
| Estimated $VO_2$max (ml/kg/min) | 25.3 (7.5) | 26.2 (9.0) | .482 |
| Timed up-and-go (s) | 6.74 (1.76) | 6.60 (1.62) | .565 |
| 10-m walk (m/s) | 1.32 (0.20) | 1.39 (0.30) | .049 |
| 400-m walk (s) | 306 (92) | 305 (71) | .902 |
| Sit-to-stand (n/30 s) | 13.9 (3.6) | 14.9 (4.2) | .077 |
| 10-m walk months (m/s) | 0.88 (0.32) | 1.03 (0.33) | .001 |
| 10-m walk numbers (m/s) | 0.94 (0.29) | 1.07 (0.30) | .003 |
| PASE score, mean (SD) | 95 (42) | 97 (49) | .789 |
| Exercise self-efficacy, mean (SD) | 16.7 (3.7) | 17.7 (3.6) | .080 |

Abbreviations: MMSE, mini mental state examination range from 0 (severe impairment) to 30 (no impairment); SDMT, Symbol Digit Modalities Test are the number of correct matches in 120 sec, with a higher score indicating a higher level of mental speed and attention; NPI, Neuropsychiatric Inventory range from 0 to 144, with higher scores indicating increased behavioral and psychological symptoms; PASE, Physical activity scale for the elderly, with higher scores indicating higher levels of habitual physical activity.

Q9 NOTE. Exercise self-efficacy range from 5 to 25, with higher scores indicating confidence in the ability to exercise in various situations.

associations between SDMT and NPI, respectively, and the included tests of physical performance or self-efficacy.

## 4. Discussion

To our knowledge, this is the first randomized controlled trial demonstrating a positive effect on cardiorespiratory fitness and exercise self-efficacy after an individualized supervised moderate-to-high–intensity aerobic exercise program for community-dwelling patients with mild AD. Furthermore, the per-protocol analysis showed that training at least twice weekly resulted in positive effects on single-task physical performance and dual-task performance.

The moderate-to-high–intensity supervised aerobic exercise program demonstrated an average increase in estimated maximal oxygen uptake corresponding to improvements found in studies on healthy middle-aged and older adults [35]. This finding demonstrates that it is feasible to implement aerobic exercise for patients with mild AD and suggests cardiovascular and metabolic benefits corresponding to that found in healthy elderly and a possibility of delaying deterioration in physical function and dependency in patients with AD [34–36]. In addition, in early stages of AD, an association between cardiorespiratory fitness and brain atrophy has been demonstrated suggesting less neurodegeneration with increasing fitness [37,38].

The exercise intervention was based on a previous study in patients with mild cognitive impairment [39] and designed to improve cardiorespiratory fitness and not overall physical performance. Most of the participants preferred to train on the ergometer bicycle, and this may explain the lack of improvement in single-task physical performance, that is, intention-to-treat results in the 400-m walk test, 10-m walk test, TUG, and STS, because these activities also require balance, coordination, and trunk stability.

Nevertheless, per-protocol analyses indicated that measures of physical performance including gait speed improve in those who exercised 2–3 times weekly.

These findings are in line with the American College of Sports Medicine's recommendation of moderate exercise for 150 min/wk or vigorous exercise for 75 min/wk or a combination of both to develop or maintain cardiorespiratory, musculoskeletal, and neuromotor fitness in healthy elderly. We had expected an improvement in the 400-m walk test as previous studies have shown significant associations between measures of maximal oxygen uptake and the 400-m walk test [28,40]. However, cognitive impairment may have increased the variance of test results in our study. In addition, our participants performed very well in all measures of single-task physical performance compared to reference values in nondemented elderly [41,42], which could have limited their response to the exercise program. This explanation is supported by a study including patients with mild-moderate dementia, showing that poorer baseline motor performance was the only factor associated with greater training response [2].

In contrast to two previous studies [10,12], we did not include any form of cognitive intervention but still found an improvement of dual-task performance in those participants who attended exercise sessions at least twice weekly. This finding is of special clinical relevance because it accentuates the role of physical activity as having the potential to benefit dual-task performance. Consequently, postponing the decline in dual-task performance by physical activity may be essential and highly relevant to activities of daily living involving gait and may thus contribute to maintain independence in patients with AD. As expected, we also found an association between change in dual-task performance and SDMT. This finding is congruent with previous cross-sectional studies in patients with all-cause dementia where dual-task performance and cognitive performance are associated [11,25,43,44].

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Table 2
Effects of the 4-month exercise intervention (intention-to-treat analyses)

| Variables | Intervention group Baseline (n = 107) | Intervention group 16 week follow-up (n = 103) | Control group Baseline (n = 93) | Control group 16 week follow-up (n = 86) | Between-group difference in mean change from baseline Δ (95% CI) | P value |
|---|---|---|---|---|---|---|
| Estimated $VO_2$max (mL/kg/min) | 25.3 (7.5) | 30.1 (7.5) | 26.2 (9.0) | 27.2 (8.7) | −4.0 (−5.8 to −2.3) | <.0001 |
| 400-m walk (s) | 306 (92.3) | 296 (79.1) | 305 (71.5) | 303 (70.5) | 9 (−2 to 20) | .118 |
| Timed up-and-go (s) | 6.74 (1.76) | 6.52 (1.72) | 6.60 (1.62) | 6.61 (1.86) | 0.24 (−0.09 to 0.56) | .151 |
| Sit-to-stand test (n/30 s) | 13.9 (3.6) | 14.3 (3.5) | 14.9 (4.2) | 15.5 (4.3) | 0.3 (−0.4 to 1.0) | .408 |
| 10-m walk (m/s) | 1.32 (0.20) | 1.34 (0.20) | 1.39 (0.30) | 1.37 (0.23) | −0.05 (−0.11 to 0.01) | .108 |
| 10-m walk months (m/s) | 0.88 (0.32) | 1.01 (0.44) | 1.03 (0.33) | 1.06 (0.32) | −0.10 (−0.20 to 0.00) | .051 |
| 10-m walk numbers (m/s) | 0.94 (0.29) | 1.01 (0.40) | 1.07 (0.30) | 1.09 (0.32) | −0.05 (−0.13 to 0.02) | .155 |
| Exercise self-efficacy score | 16.7 (3.7) | 17.4 (3.5) | 17.7 (3.6) | 16.5 (3.8) | −1.7 (−2.8 to −0.5) | .004 |

NOTE. Baseline and 16-week follow-up with unadjusted means ± SD and mean outcomes of the intervention group compared to mean outcomes in control group, beyond the difference already present at baseline, adjusted for clustering within training groups and centers. 10-m walk months (m/s): 10 m walk with naming the months backward; 10-m walk numbers (m/s): 10-m walk with counting backward from 50; Exercise self-efficacy score: range from 5 to 25, with higher scores indicating confidence in the ability to exercise in various situations.

Research has proposed a common pathway of pathologic processes in aging and dementia resulting in a decline in both cognitive and physical performance including gait [9,45–47]. Findings from studies on persons with mild cognitive impairment or dementia do to some extent support the relationship between motor performance and cognition [11,39], including a direct association between dual-task performance and gait impairment [48,49] and cardiorespiratory fitness and cognition [37,39]. However, a possible causal relationship between these functions is unknown in patients with AD.

Although we did find a markedly positive effect on cardiorespiratory fitness and physical performance in the participants with high attendance, it was not significantly related to changes in cognition. Our results are supported by findings in an exercise study including 120 older adults without cognitive impairment, which showed aerobic exercise to be effective at increasing cardiorespiratory fitness and hippocampal volume, but the change in cardiorespiratory fitness was not related to improvements in memory function [50]. Our findings are also in accordance with a cross-sectional study in AD patients where higher fitness was associated with reduced brain atrophy but not cognitive dysfunction [37].

Table 3
Effects of the 4-month exercise intervention (per-protocol analyses: > 66.6% training adherence [n = 91] vs. control)

| Variables | Between-group difference in mean change from baseline (0–16 weeks) Δ (95% CI) | P value |
|---|---|---|
| Estimated $VO_2$max (mL/kg/min) | −4.0 (−5.8 to −2.1) | <.0001 |
| 400 m walk (s) | 18 (5–31) | .007 |
| Timed up-and-go (s) | 0.47 (0.12–0.83) | .009 |
| Sit-to-stand test (n/30 s) | 0.0 (−0.8 to 0.7) | .929 |
| 10-m walk (m/s) | −0.06 (−0.12 to −0.00) | .049 |
| 10-m walk months (m/s) | −0.10 (−0.21 to 0.00) | .060 |
| 10 m walk numbers (m/s) | −0.09 (−0.17 to −0.00) | .038 |
| Exercise self-efficacy score | −1.7 (−2.8 to −0.5) | .004 |

NOTE. Mean outcomes of intervention group (n = 91) compared to mean outcomes in control group (n = 93), beyond the difference already present at baseline, adjusted for clustering within training groups and centers. 10-m walk months (m/s): 10-m walk with naming the months backward; 10-m walk numbers (m/s): 10-m walk with counting backward from 50.

NOTE. Exercise self-efficacy score: range from 5 to 25, with higher scores indicating confidence in the ability to exercise in various situations.

One explanation for the lack of correlation between improvement in cardiorespiratory fitness and cognition could be that fitness improves after a relatively short period of training, but the temporal evolution of the dose-response relationship between training and potential cognitive benefits is unknown and may in theory require a longer intervention period.

To our knowledge, this is the first study to show a significant effect on exercise self-efficacy after an exercise intervention in patients with AD. Our result is supported by findings from a group interview after a 6-month aerobic exercise program in patients with mild-moderate AD [51]. One of the themes emerging in the interviews with caregivers was "improvements in the general attitude" in the participants, which was interpreted as a result of increased self-efficacy and being more active. In addition, the participants described the exercise intervention as "socially rewarding" and a "positive experience" [51]. Anecdotally, some caregivers in our study used similar expressions when describing the impact of the exercise intervention. They reported positive changes in mood, activity level, and initiative in relation to activities. In an informal questionnaire after the intervention, the participants reported statements expressing improvements of energy and the enjoyment of exercising and being with like-minded. Based on the above, it is our perception that most of our participants and their caregivers experienced subjective positive benefits from participating in the exercise intervention, and this may have had an impact on their exercise self-efficacy. In healthy persons, higher levels of self-efficacy have been associated with both higher levels of physical activity, maintenance of physical activity, and the

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prevention of inactivity [13,14,52], all of which may potentially have a positive effect on the course of the disease.

The strength of our study includes results from a well-characterized large sample of community-dwelling patients with mild AD, a high adherence rate, and the use of outcomes validated in patients with cognitive impairment. Next, the intervention was based on the results and experiences from a pilot study in a similar group of patients with AD [18], and the exercise program was carried out rigorously and with a high intensity.

Nevertheless, the inclusion of highly well-functioning persons with AD may limit the generalizability to all patients with AD and also limited the possibility of identifying beneficial effects of exercise on some of the physical performance outcomes due to a ceiling effect. Second, including fairly physically active participants could also diminish the possibilities of finding an effect on cognitive performance and thereby detecting the association between physical performance and cognitive function. Third, the inclusion of a "usual care" control group is a limitation, as we cannot completely eliminate the effects of social interaction in the IG. Last, the intervention period was only 16 weeks, which may have limited the effects of the intervention on cognition.

## 5. Conclusion

Taken together, our results suggest that supervised physical exercise contributes to postpone the expected decline in physical activity, single-task physical performance and dual-task performance in community-living patients with mild-moderate AD. In light of the many potential benefits of exercise to community-dwelling patients with AD, future research should focus on the duration and combination of different types of exercise and its possible implications for activities of daily living, quality of life, and decreased health care cost.

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**RESEARCH IN CONTEXT** Q6

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